MATH 423: Lecture 12

Option Pricing: Preparation

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Contents

(1) Binomial tree model
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Binomial Tree Model

Recall the Binomial tree Model in Lecture 3.

For two integers $n < m$, the return over the time interval $[n, m]$ is

$$K(n,m) =: \frac{S(m) - S(n)}{S(n)} \quad \Leftrightarrow \quad S(m) = S(n)(1 + K(n,m)),$$

and one-step return is

$$K(n) := K(n-1, n) = \frac{S(n) - S(n-1)}{S(n-1)} \quad \Leftrightarrow \quad S(n) = S(n-1)(1 + K(n)).$$

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Assumption 1 (A1)

One-step returns $K(n)$, $n = 1, 2, \ldots$ are i.i.d. such that for each n,

$$K(n) = \begin{cases} u, & \text{with prob } p, \\ d, & \text{with prob } 1 - p, \end{cases}$$

where $-1 < d < u$ and $p \in (0, 1)$.

Assumption 2 (A2)

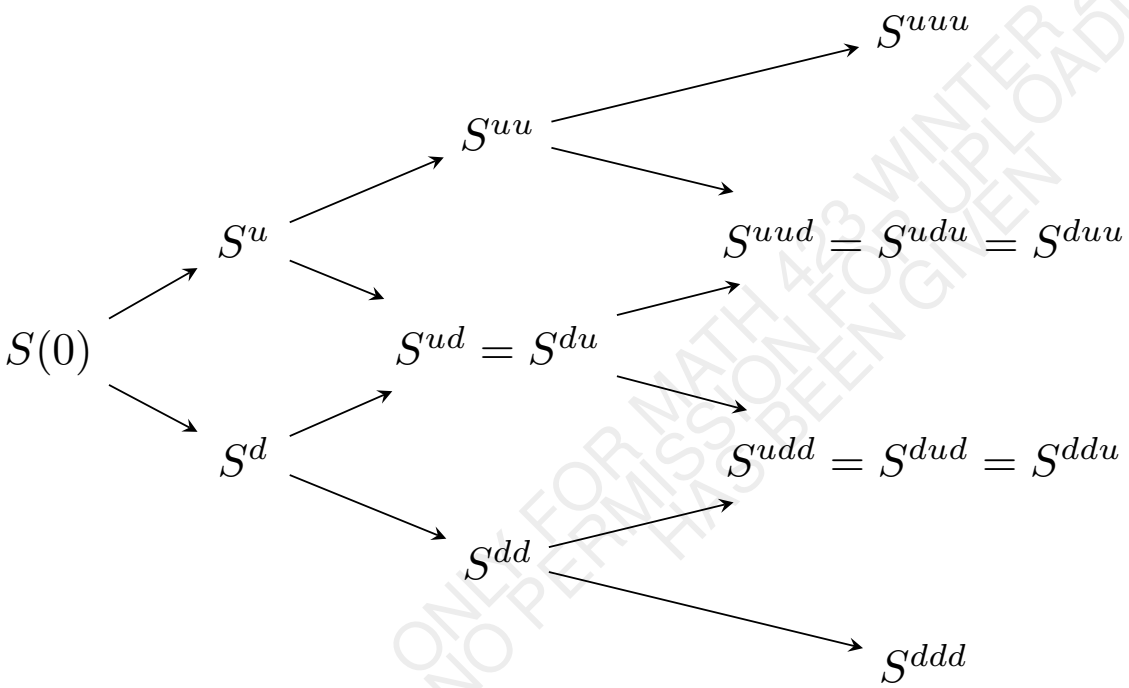
The one step return r on a risk-free investment is the same at each time step and $d < r < u$.

Binomial Tree Model

The stock price at time n could be

$$S(n) = S(0) * (1+u)^i * (1+d)^{n-i}, \qquad \text{for } i = 0, 1, \ldots, n.$$

It is illustrated by the following binomial tree:



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Under the probability measure $\mathbb{P}$, at each node, the stock price may go up with probability p, and go down with probability $1-p$ independently.

$\mathbb{P}(\text{the stock prices goes } (u,u,d)) = p * p * (1-p)$.

Binomial Tree Model, Risk-Neutral Probability

The probability of the stock price $S(n)$ taking a particular value is

$$\begin{aligned}
&\mathbb{P}(S(n) = S(0) * (1+u)^i * (1+d)^{n-i}) \\
&= \mathbb{P}(\text{the movement has } i \text{ ups and } n-i \text{ downs}) \\
&= \binom{n}{i} p^i (1-p)^{n-i}.
\end{aligned}$$

Although $S(n)$ is random, we could calculate (and are usually interested in calculating) the expected price $\mathbb{E}[S(n)]$, and compare it with the risk free investment $A(n) = A(0)(1+r)^n$.

$$\begin{aligned}
\mathbb{E}[S(n)] &= S(0)\mathbb{E}[1 + K(0,n)] \\
&= S(0)(1+\mathbb{E}[K(1)])(1+\mathbb{E}[K(1)]) \cdots (1+\mathbb{E}[K(n)]) \\
&= S(0)(1+\mathbb{E}[K(1)])^n.
\end{aligned}$$

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The comparison reduces to r v.s. $\mathbb{E}[K(1)] = pu + (1-p)d$.

What about the "ideal" situation $r = \mathbb{E}[K(1)]$?

Definition 12.1

The risk-neutral probability p^*, is such that $r = p^* u + (1 - p^*) d = \mathbb{E}^*[K(1)]$.

Remark 12.1

(a) $p^* = \dfrac{r - d}{u - d}$.

(b) The probability measure $\mathbb{P}^*$, under which the stock price goes up with probability p^* and goes down with probability $1 - p^*$ independently at each node, is called the risk-neutral probability measure.

(c) The expectation under $\mathbb{P}^*$ is called the risk-neutral expectation and is denoted by $\mathbb{E}^*$.

(d) p^* may not reflect the real/actual market probability. It is introduced for the purpose of pricing derivative securities.

(e) Since the risk-neutral probability measure is just a special choice of p $(= p^*)$, we have

$$\mathbb{E}^*[S(n)] = S(0)(1 + \mathbb{E}^*[K(1)])^n = S(0)(1 + r)^n.$$

Example 12.1

Consider a two-step binomial tee model: $S(0) = 100$, $u = 0.2$, $d = -0.2$, $r = 0.1$, then the risk-neutral probability $p^* = ?$ $\mathbb{E}^*[S(1)] = ?$ $\mathbb{E}^*[S(2)] = ?$

$p^* = \frac{r-d}{u-d} = \frac{0.1+0.2}{0.2+0.2} = \frac{3}{4}$.

$\mathbb{E}^*[S(1)] = S(0)(1 + \mathbb{E}^*[K(1)]) = S(0)(1 + r) = 100 * 1.1 = 110.$

$\mathbb{E}^*[S(2)] = S(0)(1 + \mathbb{E}^*[K(1)])^2 = S(0)(1 + r)^2 = 100 * 1.1^2 = 121.$

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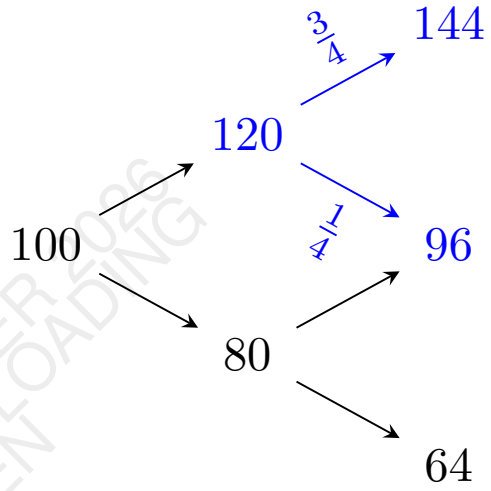
Question: In this example, after one time step, once it becomes known whether the stock price has gone up or down, what would be $\mathbb{E}^*[S(2)]$?

Martingale Property

Case 1: Suppose $S(1) = 120$.

The tree will reduce (i.e. we can erase the lower branch) to the upper branch, which is itself a one-step tree.

There are two possible values for $S(2)$, and 64 is no longer a choice.



In this case,

$$\mathbb{E}^*[S(2)] = \tfrac{3}{4} \times 144 + \tfrac{1}{4} \times 96 = 132 = 120(1+r).$$

Why does this $\mathbb{E}^*[S(2)]$ differ from the result 121 on the previous slide?

Because we are standing at $t = 1$ and find that $S(1) = 120$.

This expectation had better be rewritten as $\mathbb{E}^*[S(2)|S(1) = 120]$, the conditional expectation contingent on the fact that we know $S(1) = 120$ at $t = 1$, namely

$$\mathbb{E}^*[S(2)|S(1) = 120] = 132 = 120(1+r).$$

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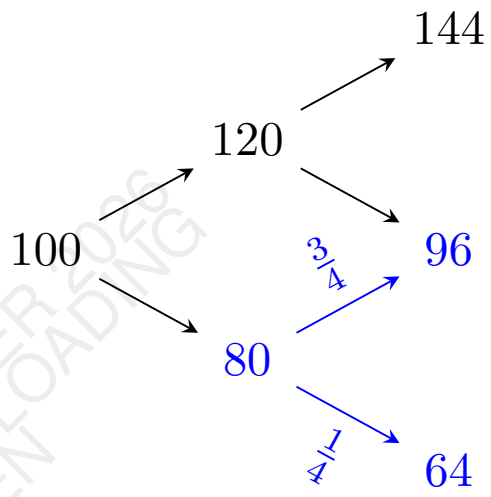
Case 2: Suppose $S(1) = 80$.

The tree will reduce (i.e. we can erase the upper branch) to the lower branch, which is itself a one-step tree.

There are two possible values for $S(2)$, and 144 is no longer a choice.

In this case, ☼



$$\mathbb{E}^*[S(2)|S(1) = 80] = \tfrac{3}{4} \times 96 + \tfrac{1}{4} \times 64 = 88 = 80(1 + r).$$

Combining these two cases, we see that

$$\mathbb{E}^*[S(2)|S(1)] = S(1)(1 + r).$$

This question and analysis can be extended to any time step in the binomial tree model: Suppose that n time steps have passed and the stock price has become $S(n)$. What is the risk-neutral expectation of the price $S(n + 1)$ after one more step?

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Proposition 12.1

Given that the stock price $S(n)$ has become known at time n, the risk-neutral conditional expectation of $S(n+1)$ will be

$$\mathbb{E}^*[S(n+1)|S(n)] = S(n)(1+r).$$

Proof:

Suppose that $S(n) = x$ after n time steps.

Then

$$\begin{tikzcd} & x(1+u) \\ S(n) = x \arrow[ur, "p^*"] \arrow[dr, "1-p^*"'] & \\ & x(1+d) \end{tikzcd}$$

$$\begin{aligned} \mathbb{E}^*[S(n+1)|S(n) = x] &= p^* x(1+u) + (1-p^*)x(1+d) \\ &= x[p^* + p^* u + (1-p^*) + (1-p^*)d] \\ &= x(1+r). \end{aligned}$$

Since this holds for every value x that $S(n)$ could take, we have

$$\mathbb{E}^*[S(n+1)|S(n)] = S(n)(1+r).$$

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Remark 12.2

The formula

$$\mathbb{E}^*[S(n+1)|S(n)] = S(n)(1+r)$$

means that standing at time step n, the risk-neutral expectation of the stock price at next stage, $S(n+1)$, is the current stock price $S(n)$ growing under the risk-free one-step interest rate.

To simplify the above expression, define the discounted stock prices

$$\tilde{S}(n) = S(n)(1+r)^{-n}.$$

Then

$$\mathbb{E}^*[\tilde{S}(n+1)|S(n)] = (1+r)^{-(n+1)}\mathbb{E}^*[S(n+1)|S(n)] = (1+r)^{-n}S(n) = \tilde{S}(n).$$

We say that the discounted stock prices $\tilde{S}(n)$ form a martingale under the risk-neutral probability measure $\mathbb{P}^*$. Therefore the probability measure $\mathbb{P}^*$ is also called the martingale measure, the probability p^* is also called the martingale probability, and the expectation $\mathbb{E}^*$ is also called the martingale expectation.

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Martingale Property

Definition 12.2

A sequence of random variables $X(0), X(1), X(2), \ldots$ such that

$$\mathbb{E}^*[X(n+1)|S(n)] = X(n)$$

for each $n = 0, 1, 2, \ldots$ is said to be a martingale with respect to $\mathbb{P}^*$.

Remark 12.3

(a) This is a special version of the definition of martingales for our course purpose.

(b) The identity $\mathbb{E}^*[\tilde{S}(n+1)|S(n)] = \tilde{S}(n)$ indeed says that the discounted stock prices $\tilde{S}(0), \tilde{S}(1), \tilde{S}(2), \ldots$ form a martingale under the risk-neutral probability measure $\mathbb{P}^*$.

(c) Since $A(n) = A(0)(1+r)^n$, the identity could be rewritten as

$$\mathbb{E}^*\left[\frac{S(n+1)}{A(n+1)} \,\middle|\, S(n)\right] = \frac{S(n)}{A(n)}.$$

(d) The result could be extended to multiple stocks, finite time steps, and finite possible stock prices.

Theorem 12.1 (Fundamental Theorem of Asset Pricing)

Suppose there are k stocks $S_1, \ldots, S_k$, and a risk-free asset A in the market. Let $S(n) = (S_1(n), \ldots, S_k(n))$ be the stock price vector and $A(n)$ be the risk-free asset price at time n. Suppose Ω is the set of all scenarios and $\mathbb{P}(\omega) > 0$ for each scenario $\omega \in \Omega$.

Then, the No-Arbitrage Principle is equivalent to the existence of a risk-neutral probability measure $\mathbb{P}^*$ on the set of scenarios Ω such that

- $\mathbb{P}^*(\omega) > 0$ for each scenario $\omega \in \Omega$;
- The discounted stock prices $\tilde{S}_i(n) = S_i(n)/A(n)$ are martingales for $i = 1, \ldots, k$:
$$\mathbb{E}^*[\tilde{S}_i(n+1)|S(n)] = \tilde{S}_i(n).$$
Here the left hand side is the conditional expectation with respect to the risk-neutral probability $\mathbb{P}^*$ computed once the stock price vector $S(n)$ becomes known at time n.

Proof not required.

Example 12.2 (Computing the risk-neutral probability $\mathbb{P}^*$)

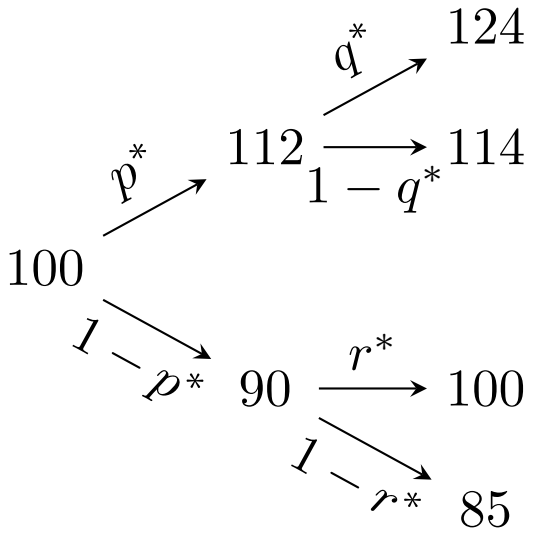
Let $A(0) = 100$, $A(1) = 110$, $A(2) = 120$. Suppose the stock price can follow the four possible scenarios (with positive probabilities). Compute the risk-neutral probability $\mathbb{P}^*(\omega_i)$ for $i = 1, 2, 3, 4$.

Scenario	$S(0)$	$S(1)$	$S(2)$
ω_1	100	112	124
ω_2	100	112	114
ω_3	100	90	100
ω_4	100	90	85

Note: This is NOT a binomial tree model because

- one-step interest rates vary over time: 0.1 vs. $1/11$.
- The u and d vary their values at different nodes: $S(2)$ has 4 different possible values.

Solution: The tree of stock prices is



So $\mathbb{P}^*$ can be represented by the branch probabilities p^*, q^*, r^* at these three nodes.

Example 12.2 continued: Theorem 12.1 for $\tilde{S}(n) = S(n)/A(n)$ can be written in the form of three equations, one for each node of the tree:

At the root, $S(0) = 100$: $\mathbb{E}^* \left[\frac{S(1)}{A(1)} \middle| S(0) = 100 \right] = \frac{100}{A(0)}$

$$\Rightarrow \frac{112p^* + 90(1-p^*)}{110} = \frac{100}{100} \Rightarrow p^* = \frac{10}{11}.$$

At the upper node at $t = 1$, $S(1) = 112$: $\mathbb{E}^* \left[\frac{S(2)}{A(2)} \middle| S(1) = 112 \right] = \frac{112}{A(1)}$

$$\Rightarrow \frac{124q^* + 114(1-q^*)}{120} = \frac{112}{110} \Rightarrow q^* = \frac{9}{11}.$$

At the lower node at $t = 1$, $S(1) = 90$: $\mathbb{E}^* \left[\frac{S(2)}{A(2)} \middle| S(1) = 90 \right] = \frac{90}{A(1)}$

$$\Rightarrow \frac{100r^* + 85(1-r^*)}{120} = \frac{90}{110} \Rightarrow r^* = \frac{29}{33}.$$

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Example 12.2 continued: For each scenario (each path through the tree) the corresponding risk-neutral probability can be computed as follows:

$\mathbb{P}^*(\omega_1) = p^* q^* = \frac{10}{11} * \frac{9}{11} = \frac{90}{121}$

$\mathbb{P}^*(\omega_2) = p^*(1 - q^*) = \frac{10}{11} * \frac{2}{11} = \frac{20}{121}$

$\mathbb{P}^*(\omega_3) = (1 - p^*)r^* = \frac{1}{11} * \frac{29}{33} = \frac{29}{363}$

$\mathbb{P}^*(\omega_4) = (1 - p^*)(1 - r^*) = \frac{1}{11} * \frac{4}{33} = \frac{4}{363}$

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We can check that these probabilities are all positive and add up to 1.

By Theorem 12.1, there is no arbitrage.

Eg 12.2 alternative approach: instead of finding p^*, q^*, r^*, we could directly set up equations (based on Theorem 12.1) to find $p_i^* := \mathbb{P}^*(\omega_i)$, $i = 1, 2, 3, 4$.

At the root, $S(0) = 100$: $\mathbb{E}^* \left[\frac{S(1)}{A(1)} \middle| S(0) = 100 \right] = \frac{100}{A(0)}$

$$\Rightarrow \frac{112}{110} p_1^* + \frac{112}{110} p_2^* + \frac{90}{110} p_3^* + \frac{90}{110} p_4^* = \frac{100}{100}.$$

At the upper node at $t = 1$, $S(1) = 112$: $\mathbb{E}^* \left[\frac{S(2)}{A(2)} \middle| S(1) = 112 \right] = \frac{112}{A(1)}$

$$\Rightarrow \frac{124}{120} \frac{p_1^*}{p_1^* + p_2^*} + \frac{114}{120} \frac{p_2^*}{p_1^* + p_2^*} = \frac{112}{110}.$$

At the lower node at $t = 1$, $S(1) = 90$: $\mathbb{E}^* \left[\frac{S(2)}{A(2)} \middle| S(1) = 90 \right] = \frac{90}{A(1)}$

$$\Rightarrow \frac{100}{120} \frac{p_3^*}{p_3^* + p_4^*} + \frac{85}{120} \frac{p_4^*}{p_3^* + p_4^*} = \frac{90}{110}.$$

In addition, the probabilities should add up to 1:

$$p_1^* + p_2^* + p_3^* + p_4^* = 1.$$

Solve these four equations to get

$$p_1^* = \frac{90}{121}, \quad p_2^* = \frac{20}{121}, \quad p_3^* = \frac{29}{363}, \quad p_4^* = \frac{4}{363}.$$

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